

Nicole Hee-Yeon Kim

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Nationality: United States of America and Republic of Korea (Dual Citizenship)

RESEARCH INTEREST

Efficient and Scalable AI Systems; Optimization for Large-Scale Deep Learning Models; Training and Inference Efficiency in LLMs and VLMs; Robust and Reliable AI

EDUCATION

- **Korea Advanced Institute of Science and Technology** Feb 2024 - Feb 2026
Master's Program, Department of Industrial & Systems Engineering, Overall GPA: 3.85/4.0 Daejeon, South Korea
- **Yonsei University** Mar 2019 - Feb 2024
Bachelor's Program, Department of Industrial Engineering, Overall GPA: 3.70/4.0 Seoul, South Korea

RESEARCH EXPERIENCE

- **DISLab, Department of Industrial & Systems Engineering, KAIST** Feb 2024 - Feb 2026
Master's Student Daejeon, South Korea
 - Led the research project "Robust Dataset Condensation using Supervised Contrastive Learning" as the sole first author (accepted at ICCV 2025).
 - Led the research project "Qualcomm Industry-Academia Collaboration Project" titled "How to Represent and What to Retrieve: Adaptive Planning for Agentic Personal Assistants".
 - Participated in research project "Development of VLM-based PDF Document Understanding and Automated Road Alignment Coordinate Extraction Technology", supported by ETRI.
 - Participated in research project "Enhancing AI Model Reliability Through Domain-Specific Automated Value Alignment Assessment".
 - Participated in research project "Naver Cloud Consortium: Omni Foundation Model Project", supported by the Ministry of Science and ICT, NIPA, NIA, and IITP.
- **MLAI Lab, Department of Applied Statistics, Yonsei University** Sep 2023 - Feb 2024
Undergraduate Research Intern Seoul, South Korea
 - Designed and constructed the PlayingCard-10 dataset, a core component of "IMC: A Benchmark for Invariant Learning under Multiple Causes" (CVPR 2025 Workshop Best Paper Award).

PUBLICATIONS

1. **Kim, N.** and Song, H. (2025). Robust Dataset Condensation using Supervised Contrastive Learning. In *Proceedings of International Conference on Computer Vision (ICCV 2025)*, Main Conference Paper.
2. **Kim, N.**, Choi, J., Lee, Y., Song, H. (2025). Robust Dataset Condensation via Semi-Supervised Learning. In *Proceedings of the Korea Computer Congress (KCC 2025)*, Selected for Oral Presentation.
3. **Kim, N.**, Lee, Y., Song, H. (2024). Robust Dataset Condensation via Supervised Contrastive Learning. In *Proceedings of the Korea Software Congress (KSC 2024)*, Selected for Oral Presentation.
4. Ban, M., Choi J., Min H., **Kim, N.**, Kim M., Lee J., Song H. Completing Missing Annotation: Multi-Agent Debate for Accurate and Scalable Relevant Assessment for IR Benchmarks. In *Proceedings of International Conference on Learning Representations (ICLR 2026)*.
5. Lee, Y., Deng, J., **Kim, N.**, Min, H., Yun, T., Ban, M., Song, H. (2025). Towards a Holistic and Automated Evaluation Framework for Multi-Level Comprehension of LLMs in Book-Length Contexts. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*, Main Conference Paper.
6. Min, H., Lee, Y., Ban, M., Deng, J., **Kim, N.**, Yun, T., Su, H., Cai, J., Song, H. (2025). Towards Multi-dimensional Evaluation of LLM Summarization across Domains and Languages. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, Main Conference Paper.

7. Kim, T., Lee, S., Kang, J., Choi, Y., Yun, W., **Kim, N.**, Chen, Z., Xie, L., Song, K. (2025). IMC: A Benchmark for Invariant Learning under Multiple Causes. In *Proceedings of the CVPR 2025 Workshop on Domain Generalization: Evolution, Breakthroughs, and Future Horizons*, Best Paper Award.

8. Oh, J., Choi, J., **Kim, N.**, Yun, T., Kwon, R., Song, H. (2025). Learning to Verify Summary Facts with Fine-Grained LLM Feedback. In *Proceedings of the 2025 International Conference on Computational Linguistics (COLING 2025)*, Selected for Oral Presentation.

9. Oh, J., Choi, J., **Kim, N.**, Song, H. (2025). Improving Language Model Quality through LLM-based Fine-Grained Hallucinated Summary Generation. *Journal of Computing Practice*, vol. 31(2), pp. 91-97.

10. Oh, J., Choi, J., **Kim, N.**, Song, H. (2024). Improving the Text Summary Quality Through Understanding the Hallucination Level of Summarization Using Large Language Models. In *Proceedings of the Korea Computer Congress (KCC 2024)*, Selected for Oral Presentation.

PATENT

- Robust Dataset Condensation using Supervised Contrastive Learning (2025, Pending).
- Cost-efficient and Accurate Debate-based Relevance Assessment System with Multi-agents(2025, Pending).

SCHOLORSHIPS

• KAIST Support Scholarship Government-funded full tuition scholarship for M.S. program	Feb 2024 - Feb 2026 KAIST
• Brain Korea 21 (BK21) Scholarship Government-funded research scholarship for graduate students	Feb 2024 - Feb 2026 National Research Foundation of Korea
• Yonsei Welfare Scholarship Full tuition scholarship for B.S. program	Mar 2019 - Jun 2023 Yonsei University
• University Innovation Support Scholarship Scholarship awarded for the development and advancement of an innovative start-up idea	Sep 2020 - Feb 2021 Yonsei University

HONORS AND AWARDS

• Best Paper Award CVPR 2025 Workshop on Domain Generalization: Evolution, Breakthroughs, and Future Horizons	Jun 2025
• Outstanding Presentation Paper Award Korea Computer Congress 2024	Jul 2024
• 1st Place, Promotional Video Contest Department of Industrial Engineering, Yonsei University	Feb 2022
• Yonsei Social Entrepreneurship Award Yonsei University	Feb 2021

LANGUAGE PROFICIENCY

- **Languages:**Korean (Native), English (Fluent, TOEFL iBT: 111 [R30 L27 S28 W26])

TEACHING EXPERIENCE

• Yonsei-NAVER Cloud Data Science Program Assisted in designing, editing, and organizing instructional materials and coding sessions	Dec 2020 - Dec 2023 Yonsei University
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LEADERSHIP EXPERIENCE

• Academic & Campus Life Mentor for Freshmen Institute for Higher Education Innovation, Yonsei University	Oct 2019 - Aug 2020
• Freshman Class President Department of Industrial Engineering, Yonsei University	Mar 2019 - Aug 2019